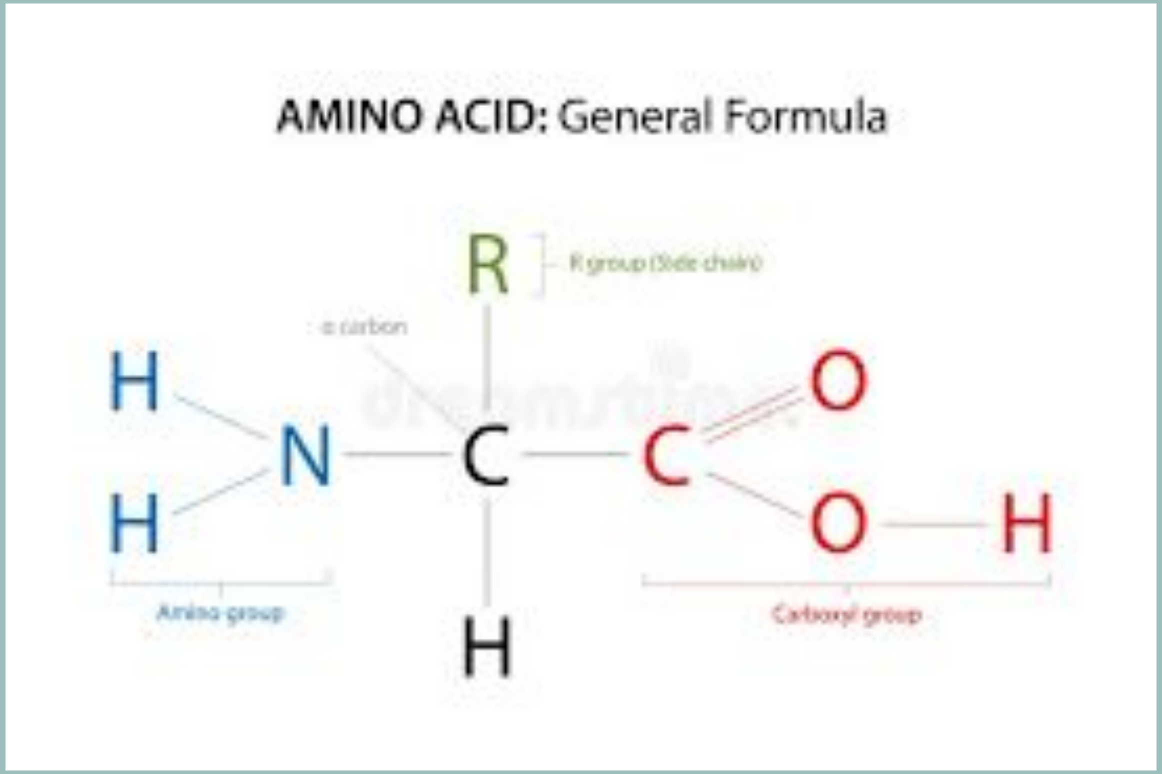
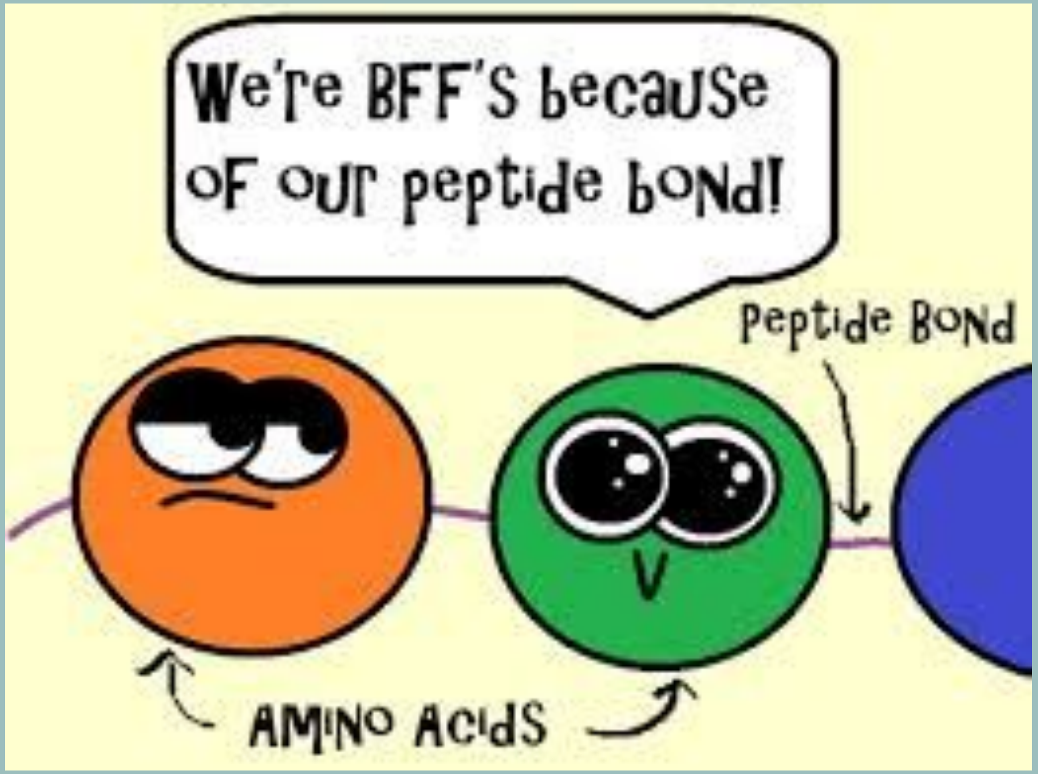




IMPORTANCE OF AMINO ACIDS & ROLE OF PROTEINS IN HEALTH & DISEASE

DR WAJEEHA SHAMS ANSARI

LEARNING OBJECTIVES

At the end of this lecture, students should be able to:

* Classify amino acids

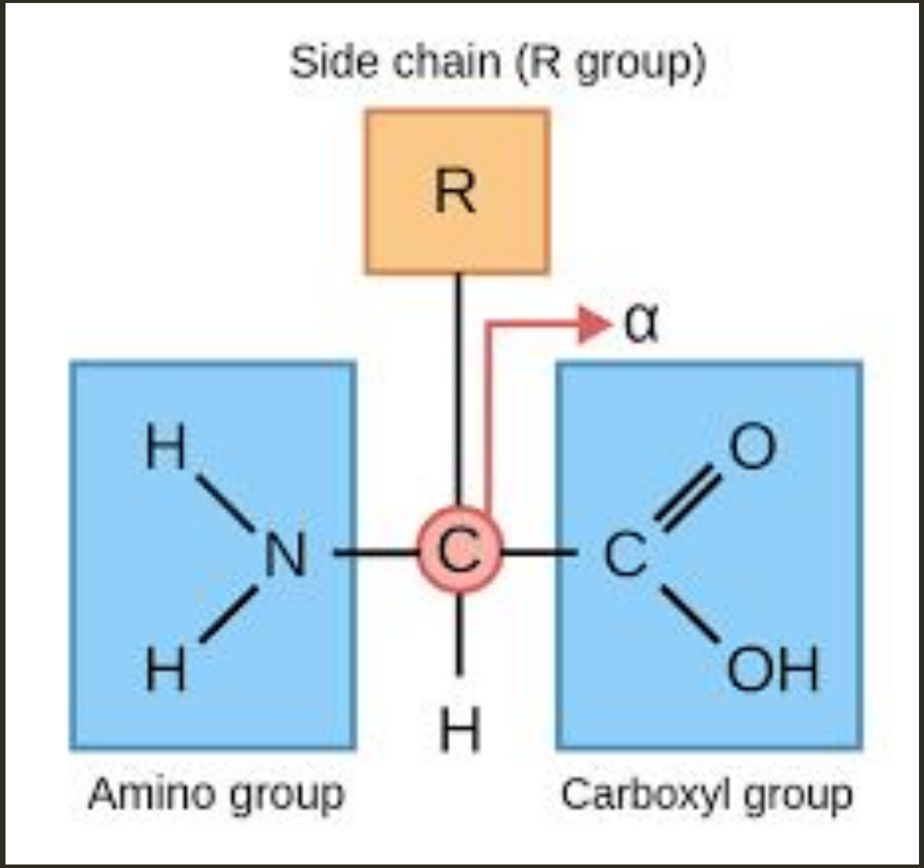
* Explain their biological importance

* Describe inborn errors of metabolism

* Explain role of protein in health & disease

STRUCTURE OF AMINO ACIDS

* Central α-carbon
* Amino group (-NH2)
* Carboxyl group (-COOH)
* Hydrogen
* Variable R group



CLASSIFICATION (NUTRITIONAL)

Essential

- Leucine
- Isoleucine
- Lysine
- Methionine, etc.

Non-Essential

- Alanine
- Aspartate
- Glutamate

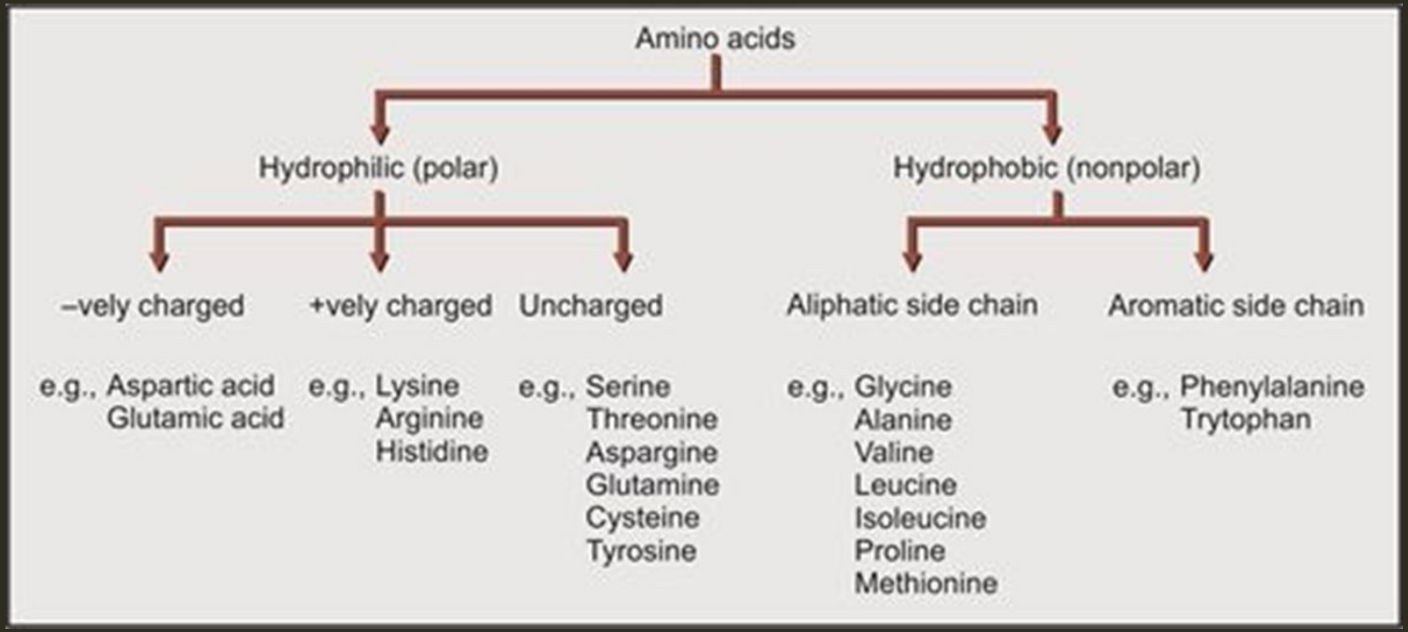
Conditionally Essential

- Arginine

THE 11 NON-ESSENTIAL AMINO ACIDS	THE 9 ESSENTIAL AMINO ACIDS
Alanine	Histidine
Arginine	Isoleucine
Asparaginee	Leucine
Aspartic acid	Lysine
Cysteine	Methionine
Glutamic acid	Phenylalanine
Glutamine	Threonine
Glycine	Tryptophan
Proline	Valine
Serine	
Tyrosine	

CLASSIFICATION (SIDE CHAIN)

- Nonpolar
- Polar uncharged
- Acidic (Aspartate, Glutamate)
- Basic (Lysine, Arginine)



BIOLOGICAL IMPORTANCE

Amino acids are:

- Building blocks of proteins
- Energy sources
- Nitrogen carriers
- Precursors of bioactive molecules

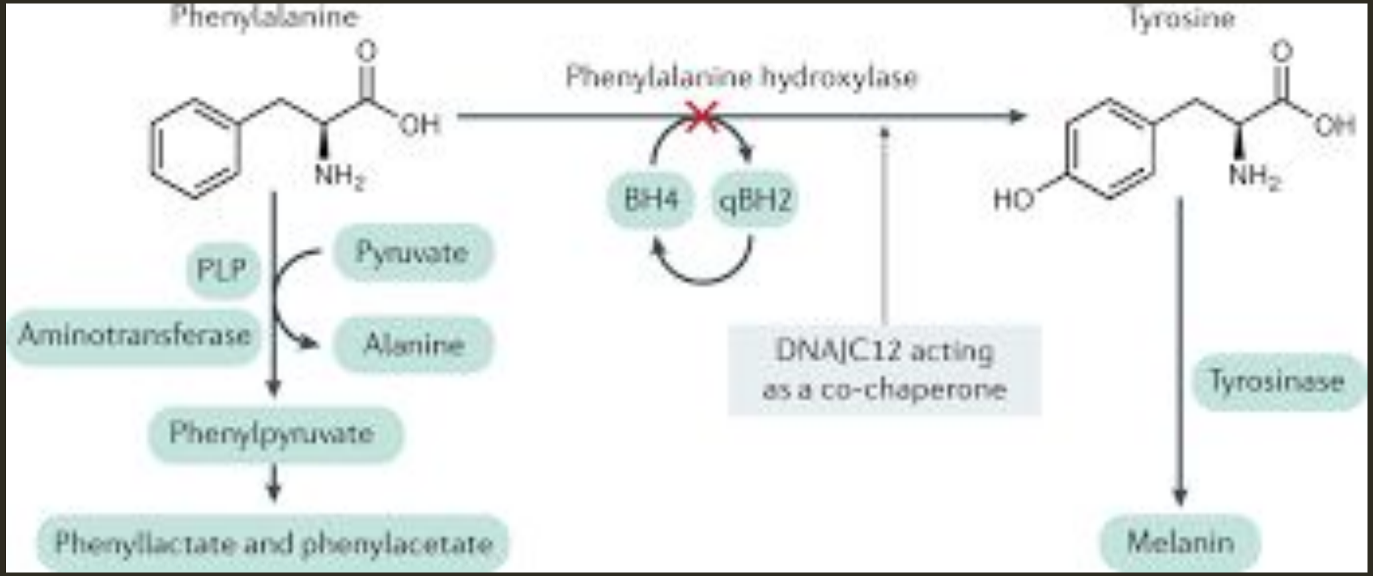
AMINO ACIDS AS PRECURSORS

Amino Acid		Product
Tyrosine	→	Dopamine, Thyroxine
Tryptophan	→	Serotonin, Niacin
Glycine	→	Heme
Glutamate	→	GABA
Arginine	→	Nitric Oxide

AMINO ACIDS DISORDERS

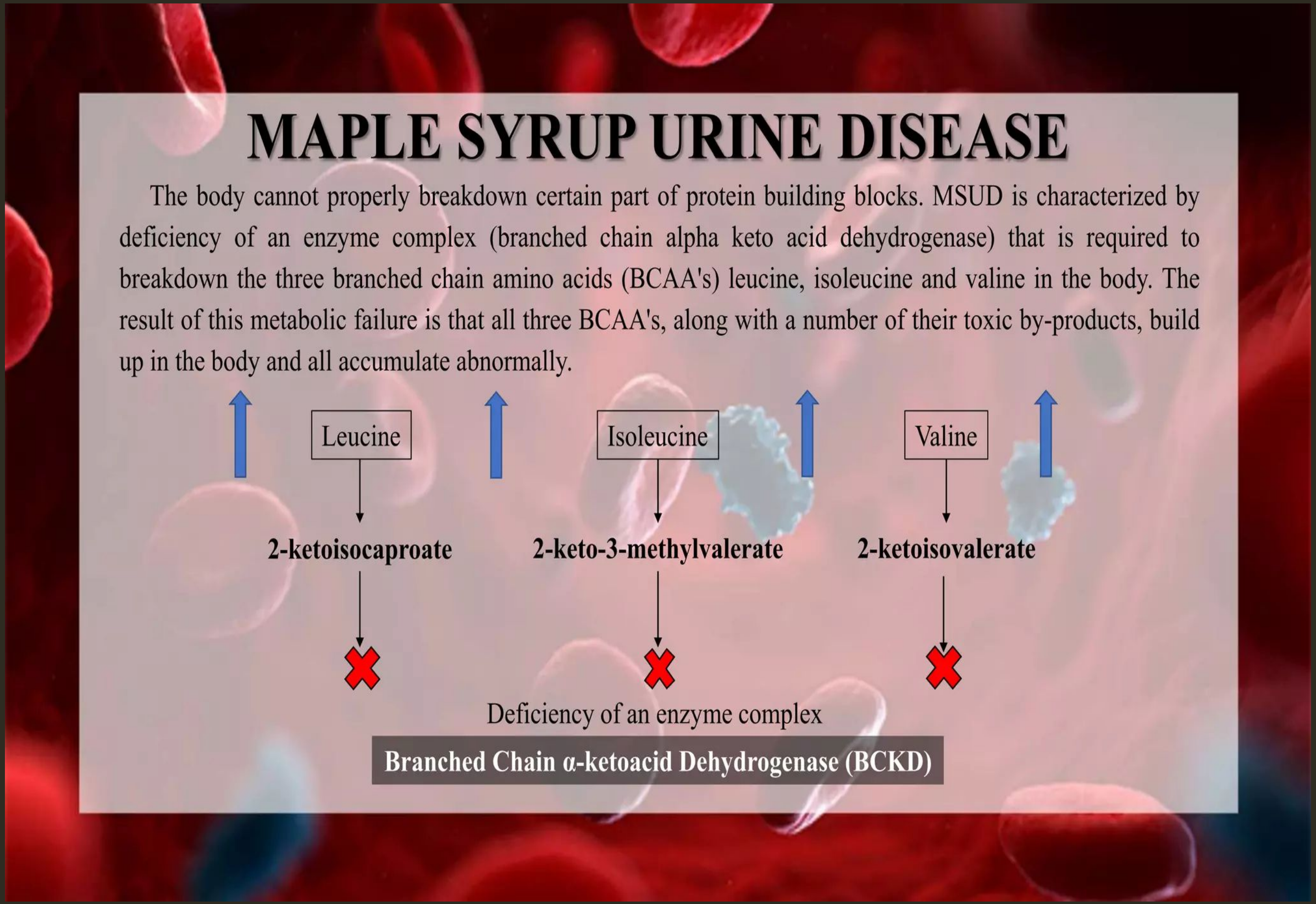
PHENYLKETONURIA

* Deficiency: Phenylalanine hydroxylase
* Phenylalanine ⬆
* Mental retardation
* Musty odor
* Treatment: Low Phenylalanine diet



MAPLE SYRUP URINE DISEASE

* Defect in BCAA metabolism
* Sweet smelling urine
* Severe neurological damage



OTHER DISORDERS

Homocystinuria

* Lens dislocation

* Thrombosis

Alkaptonuria

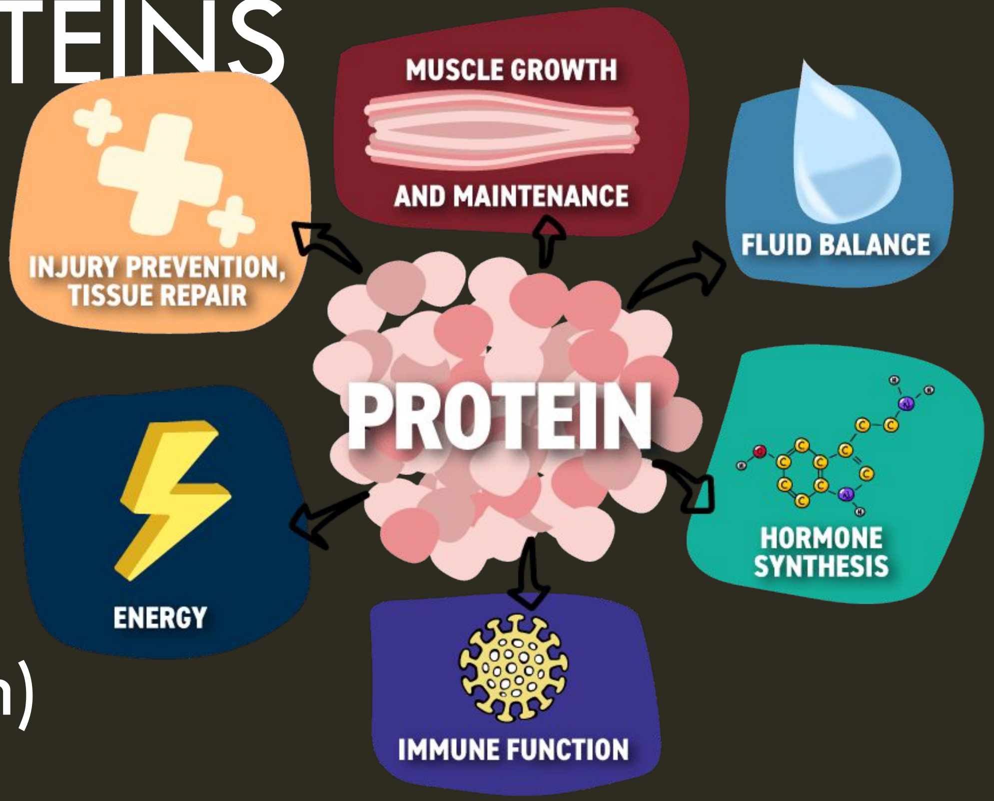
* Dark urine

* Ochronosis

PROTEINS

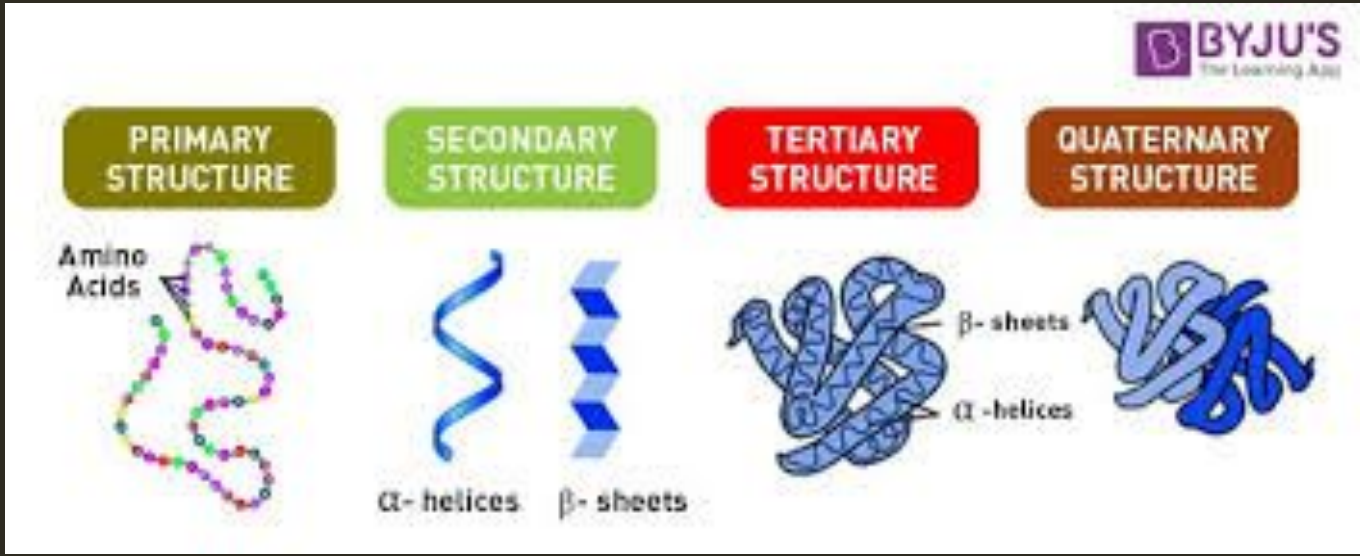
FUNCTIONS OF PROTEINS

* Structural (Collagen)
* Enzymes
* Hormones (Insulin)
* Transport (Hemoglobin, Albumin)
* Immunity (Immunoglobulins)



LEVELS OF PROTEIN STRUCTURE

1. Primary - Amino acid sequence
2. Secondary - a-helix, B-sheet
3. Tertiary - 3D folding
4. Quaternary - Multiple subunits



PROTEIN DENATURATION

Causes:

* Heat

* pH change

* Heavy metals

Effects:

* Loss of function

* Example: Oral burns

PROTEIN ENERGY MALNUTRITION

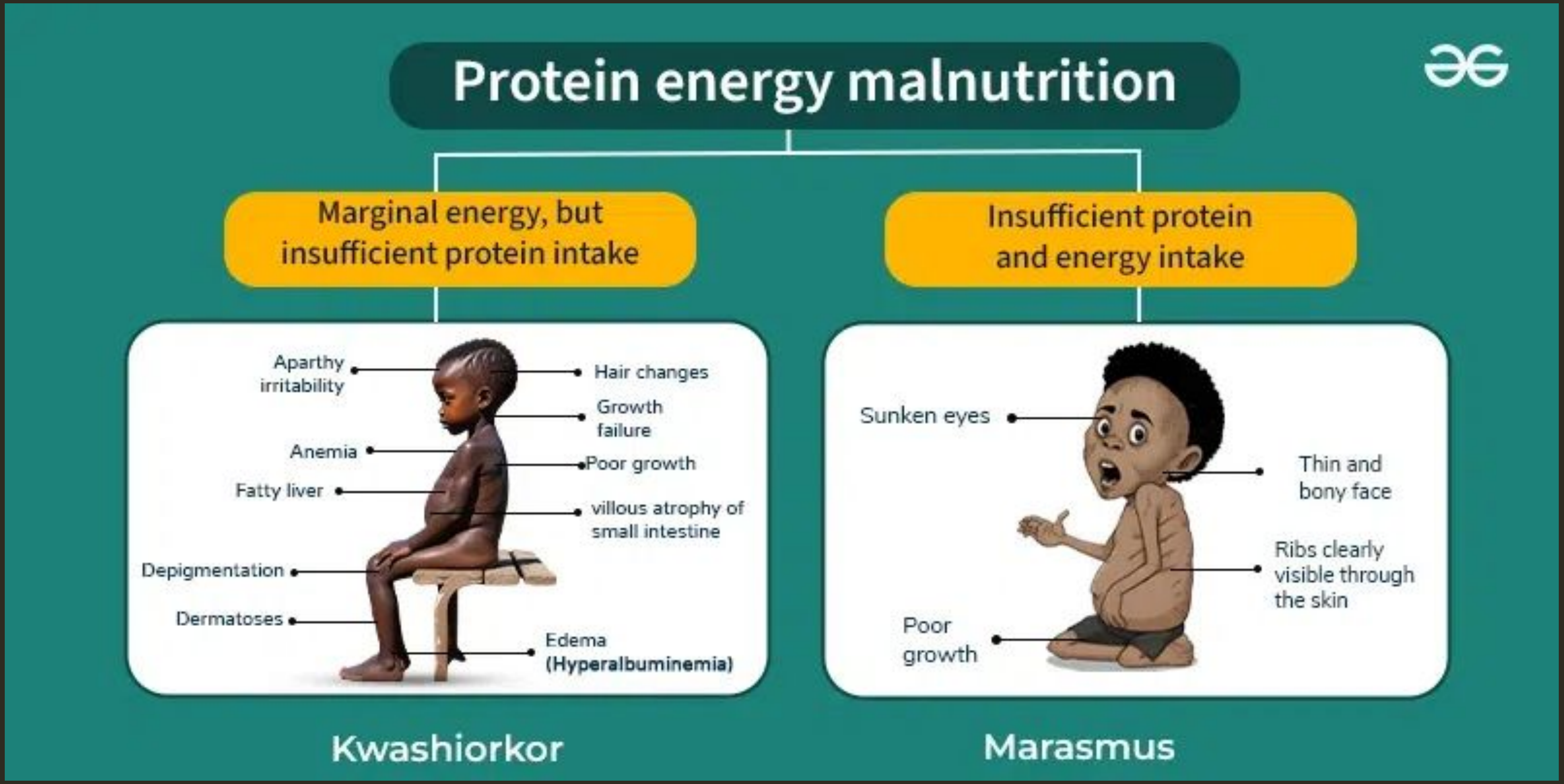
Kwashiorkor

* Edema

* Fatty liver

Marasmus

* Severe wasting

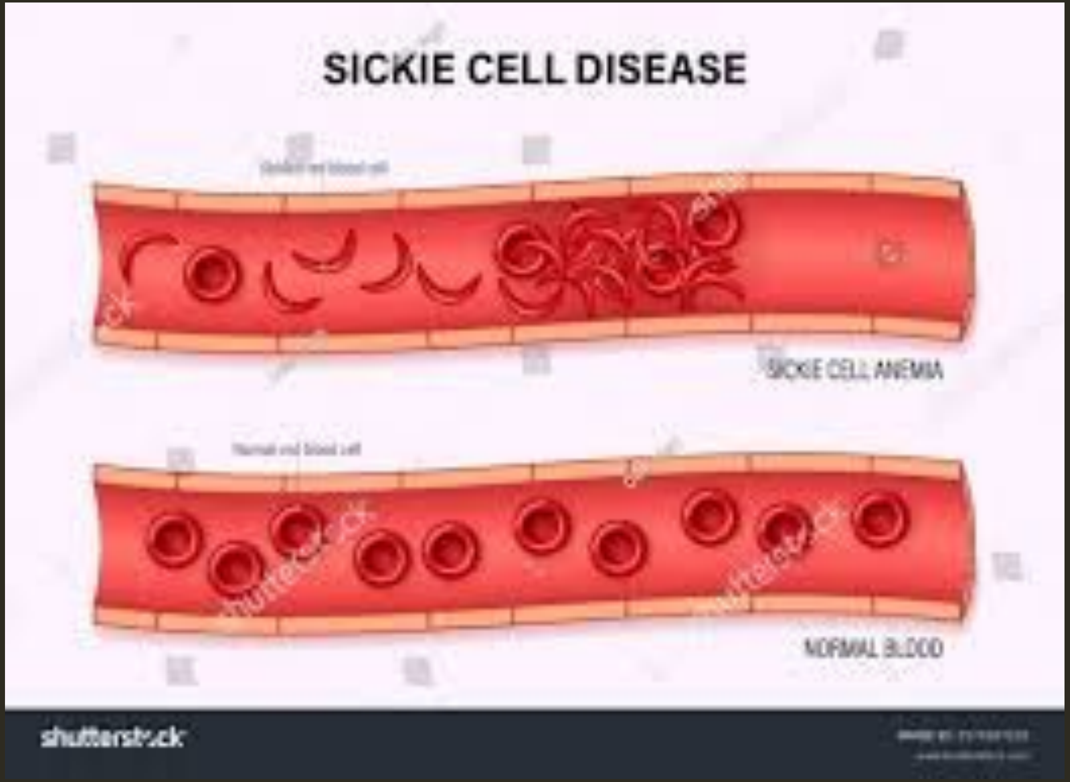


STRUCTURAL PROTEIN DISORDERS

Sickle Cell Anemia

* Mutation in B-globin

* Valine replaces glutamate



COLLAGEN DISORDERS

Scurvy

* Bleeding gums

* Poor healing

Osteogenesis Imperfecta

* Brittle bones

* Dentinogenesis imperfecta

PLASMA PROTEINS

* Albumin → Oncotic pressure
* Acute phase proteins → Inflammation
* Globulins → Immunity

CLINICAL IMPORTANCE FOR DENTISTS

* Collagen → Periodontal health

* Protein deficiency → Delayed healing

* Scurvy → Bleeding gums

* Genetic disorders → Oral manifestations

BIOLOGIC SIGNIFICANCE OF PROTEINS

BALANCE DIET / DIET :

A one that contains all the five types of dietary ingredients i:e Proteins , fats , carbohydrates , vitamins and minerals in amounts sufficient for the particular individual upon his age and gender etc.

PROTEINS :

Humans have no dietary requirements for protein per gender rather the protein in food provides essential amino acids.

BIOLOGIC VALUE OF PROTEINS:

"The biologic value of a dietary protein is a measure of its quality , that is , its ability to provide the essential amino acids required for tissue maintenance."

Biologic value of proteins from animal sources:

Proteins from animal sources Eg: meat , poultry , milk and fish have a high biologic value b/c they contain all the essential amino acids in proportions similar to those required for synthesis of human tissue proteins.

Biologic value of proteins from plant sources :

Proteins from plant sources Eg: wheat , corn , rice and beans have a lower biologic value than animal proteins. However proteins from different plant sources may be combined in such a way that the result is equivalent in nutritional value to animal protein .

SUMMARY

The standard daily protein requirement for a sedentary adult is a minimum of **0.8 grams per kilogram (g/kg) of body weight** (or 0.36g per pound) to avoid deficiency.

Following is the summary of important roles of proteins :

- Formation of protoplasm
- Nucleoproteins are made up of proteins act as heredity carriers
- Enzymes are proteins
- Proteins are an integral part of viruses

CONTINUED…

- Each one gram of dietary proteins furnishes 4.1kilocalories when oxidized in the body.
- Hemoglobin acts as a carrier of oxygen.
- Some act as hormones.
- The proteins present in blood plasma act as colloidal particles and exert an osmotic pressure.
- Proteins make complexes with carbohydrates and lipids.


K Thanks
Bye!